

OPERATING SYSTEMS

Process Management and Scheduling

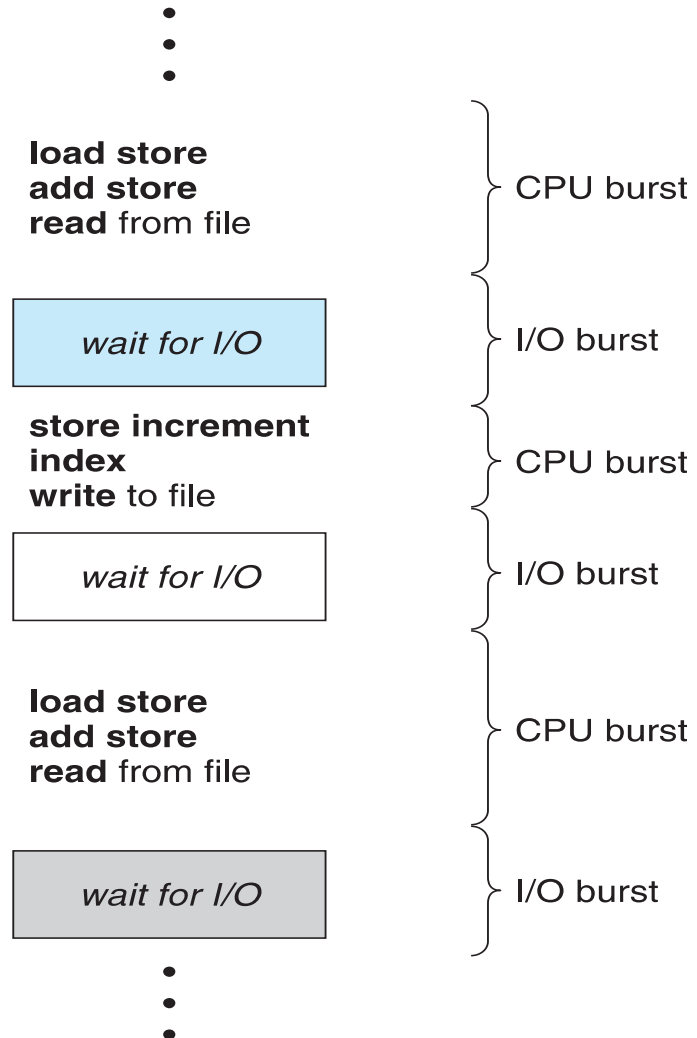
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- Maximum CPU utilization obtained with multiprogramming
 - waiting for I/O is wasteful
 - 1 thread will utilize only 1 core
- CPU–I/O Burst Cycle
 - Process execution consists of:
 - a **cycle** of CPU execution
 - and I/O wait
- **CPU burst** followed by **I/O burst**
- CPU burst distribution is of main concern

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Process Scheduling - Basic Concepts



- **Short-term scheduler**
 - Selects 1 process from the ready queue then allocates the CPU to it
 - Queue may be ordered in various ways : FIFO Queue, Priority Queue, tree, linked list

- **Dispatcher**
 - A module that gives control of the CPU to the process selected by the short-term scheduler; this involves:
 - switching context
 - switching to user mode
 - jumping to the proper location in the user program to restart that program

- **Dispatch latency**
 - Time it takes for the dispatcher to stop one process and start another running
 - This time should be as small as possible
 - problems that can arise
 - Access to shared data – race condition can happen, if not handled

- CPU scheduling decisions may take place when a process:
 1. Switches from running to waiting state
 2. Switches from running to ready state
 3. Switches from waiting to ready
 4. Terminates

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Process Scheduling - Basic Concepts

- Scheduling under 1 and 4 is called **nonpreemptive (=cooperative)**
- All other scheduling is called **preemptive**
 - Process can be **interrupted** and must release the CPU
 - Special care should be taken to prevent problems that can arise
 - Access to shared data – race condition can happen, if not handled



- How do we decide which scheduling algorithm is good?
- Many **criteria** for judging this has been suggested
 - Which characteristics considered can **change significantly** which algorithm is considered the best

- **CPU utilization** – keep the CPU as busy as possible
- **Throughput** – # of processes that complete their execution per time unit
- **Turnaround time** – amount of time to execute a particular process
- **Waiting time** – amount of time a process has been waiting in the ready queue
- **Response time** – amount of time it takes to start responding
 - Used for interactive systems
 - Time from when a request was submitted until the first response is produced

- Scheduling algorithm optimization criteria
 - Max CPU utilization
 - Max throughput
 - Min turnaround time
 - Min waiting time
 - Min response time



THANK YOU

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